

h_da
HOCHSCHULE DARMSTADT
UNIVERSITY OF APPLIED SCIENCES
fbeit
FACHBEREICH ELEKTROTECHNIK
UND INFORMATIONSTECHNIK

Power System Operation - Lab

Prof. Dr. Sebastian Weck, Lars Weispfenning

WiSe 2023/2024

h_da
HOCHSCHULE DARMSTADT
UNIVERSITY OF APPLIED SCIENCES
fbeit
FACHBEREICH ELEKTROTECHNIK
UND INFORMATIONSTECHNIK



Session 1

Exercises 1-5

Power System Operation - Lab

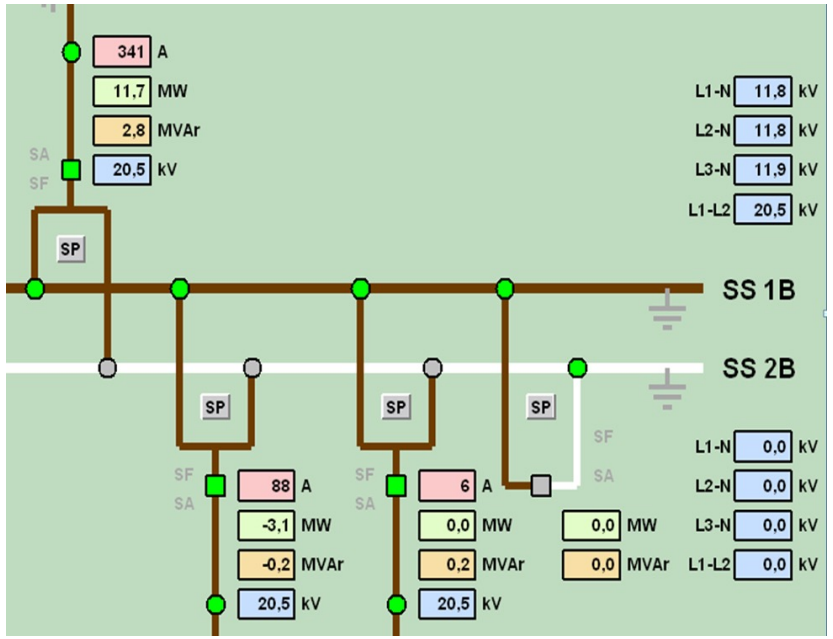
Exercises Overview

- Class A: Introductory exercises
  - Menu bar, network diagrams, operation, alarm guidance, protocols, curve displays
  - topology, measurements
- Class B: Normal operation
  - Grid changeovers, disconnections, busbar changes
  - Switching, interlocking and switching calls
  - Adjustment of earth fault quenching coil
  - Load control, demand side management
  - Voltage regulation, setpoint specifications. Control of transformer stages
- Class C: Equipment and systems in the limit range
  - Load flow control
  - Circuit currents with parallel transformers
  - Island grids in the limit range
  - Wind farm - power fluctuations and voltage values
- Class D: Fault Situations
  - Ground fault and ground fault search, ground fault relay, double ground fault
  - Short-circuit and short-circuit search, protective excitations
  - Regulating transformers and parallel connection, circuit currents
  - Power plant failure and frequency control, overload with frequency collapse, frequency protection, Q-U protection, feed-in and generation management
  - Effect of primary and secondary control
- Class E: Economic indicators and optimizations in the control system
  - Grid losses and loss costs
  - Emissions from electricity production
  - Transmission revenues
  - Outage costs during shutdowns

Power System Operation - Lab

Exercise - in advance

- Important: Uniform language!
  - Information and switching talks
    - Information talks for an exchange of information
    - Switching calls for switching orders by radio to on-site personnel
  - **DA** handover (disposition authorization)
- Switching devices/circuits
  - "**ON - OFF**": Switching for circuit breakers and switch-disconnectors that can can interrupt typical operating currents
  - "**Open - Close**": for disconnectors that switch only load currents
  - "**Disconnect**": Isolating distance to live power supplies on all sides and all poles and all poles (work safety - backfire safety)
  - "**Ready for operation**": creates the required connection via all necessary disconnectors (on), the circuit breaker(s) is/are still switched off

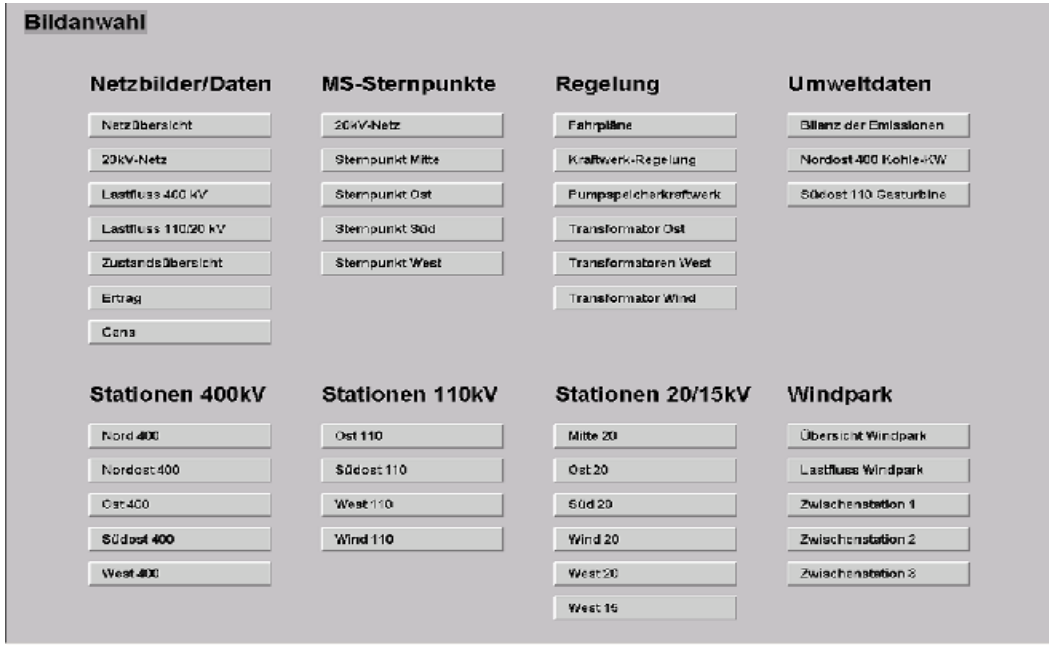


Power System Operation - Lab

Exercise 1

1a) Explore menu bar





Menu item
"Book"
provides
Selection list
of the grid images

1b) Become familiar with grid pictures and control system operation

Power System Operation - Lab

Exercise 2

a) Determine from the screen "Zustandsübersicht"

- Current network loading : ..... MW
- Current network losses : .....MW
- Cost of losses per hour : .....Euro/h

b) Determine from the screen "Lastfluss 400 kV":

- For the coal power plant in „Nordost 400"
  - Active power P: .....MW (+ is feeding)
  - Reactive power Q: ..... MVAr (+ is feeding inductively)
- For the line „West 400 - Nord 400"
  - Loading (%) : .....
  - Active power losses ΔP : .....MW
  - Reactive power generation/consumption ΔQ : .....MVAr

  (ΔQ is the sum of both reactive power measurements. $\Delta Q < 0$: consumption of inductive reactive power)

c) Determine from the screen "West 20"

- Active and reactive power and the current in the branch to the corridor "Festwiese": P = ...MW, Q= .....MVAr, Current I = ...A
- Busbar voltage for SS1 in West 20: U_{1E}=..... U_{2E}=..... U_{3E}=..... U_{12}=..... kV

Power System Operation - Lab

Exercise 3

Switching a line for circuit breaker maintenance:

- In Area „Mitte 20“ the circuit breaker of the branch to West 20 shall be maintained.
- Check if the switching bay is critically needed. If yes, switch to an alternative line, else switch off.
- Perform switching according to the 5 security rules! What is to be done centrally, what locally?
  - Switching: Open circuit breaker and open the disconnector on all sides.
  - Protect against reconnection: Set work markers („Arbeitsmarkierungen“) of type danger to all relevant switch components: „Revision CB by Müller till 2pm“
  - Check for zero voltage
  - Grounding
  - Covering of near devices under voltage
- Transfer disposition authorization (DA) from the control center to local work responsible (WR).
- The WR Reports the completion of the work.
- Take DA and restore the original state.

Power System Operation - Lab

Exercise 4

Bubar isolation in „UW Ost 20“

- In „UW OST 20“, the 20 kV SS1 (SS, Sammelschiene = bus bar) shall be isolated and replaced by bus bar 2.
- The transformer feed and all junctions shall be transferred to SS2 without disruption. Strategy:
  - First close the busbar coupler
  - All branches of the „OST 20“ system must be switched from SS1 to SS2 "in sequence" without interruption.
  - The disconnector bridge always exists temporarily only
  - The transformer feed is switched, when about half of the load has been transferred.
  - Last, open the busbar coupler and the disconnectors, so the busbar is isolated.
- 2nd group member: Transfer all branches of substation OST-20 from SS2 back to SS1 without disruption
- 3rd group member:: reperform busbar isolation, so every one has done one iteration.

Power System Operation - Lab

Exercise 5

West 20 – Transformer switch and authorization transfer

- In „West 20“, the conducting transformer shall be maintained by the grid service and shall be replaced disruption-free by the parallel transformer. The earthing coils shall be switched as well.
- Procedure (5 security rules shall be applied):
  - Set Voltage control of TR69 and TR70 to manual („Automatik AUS“)
  - Manually set voltages by increasing/decreasing the steps
  - Switch transformers to parallel operation (
  - Connect transformers in parallel (Condition: Voltage difference < 300V)
  - Switch earthing coils to the operating transformer
    - Select graphic earthing coil
    - Close the second busbar coil disconnector
    - Close the second starpoint main disconnector
    - Open the first starpoint main disconnector
    - Open the first busbar coil disconnector
  - Isolate , secure (marker „Gefahr“), ground TR 70
  - Transfer authorization to the Asset Management Service Team
- After the end of maintenance, re-establish authorization for the transformer
- Next group member, as above…

